

Stochastic Financial Models

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Third Year

1 Martingales

We work with a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

1.1 Definitions

A *filtration* is a sequence of σ -algebras $(\mathcal{F}_n)_{n \geq 0}$ such that

$$\mathcal{F}_n \subseteq \mathcal{F}_{n+1} \subseteq \mathcal{F}.$$

A *random process* is a sequence $X = (X_n)_{n \geq 0}$ of random variables.

Given a filtration $(\mathcal{F}_n)_{n \geq 0}$ on $(\Omega, \mathcal{F}, \mathbb{P})$, we say that X is a *martingale* if:

- (a) X is adapted, i.e. X_n is $\mathcal{F}_n$ -measurable for all n .
- (b) X is integrable, i.e. X_n is integrable for all n :

$$\mathbb{E}[|X_n|] < \infty.$$

- (c) X satisfies the martingale property:

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n \quad \text{a.s., for all } n.$$

Given (a) and (b), condition (c) is equivalent to

$$\mathbb{E}[X_{n+1} \mathbf{1}_A] = \mathbb{E}[X_n \mathbf{1}_A] \quad \text{for all } n \text{ and all } A \in \mathcal{F}_n.$$

If (c) is replaced by

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n \quad \text{a.s., for all } n,$$

or, equivalently,

$$\mathbb{E}[X_{n+1} \mathbf{1}_A] \geq \mathbb{E}[X_n \mathbf{1}_A] \quad \text{for all } n \text{ and all } A \in \mathcal{F}_n,$$

then $(X_n)_{n \geq 0}$ is a *submartingale*.

If (c) is replaced by

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n \quad \text{a.s., for all } n,$$

or, equivalently,

$$\mathbb{E}[X_{n+1} \mathbf{1}_A] \leq \mathbb{E}[X_n \mathbf{1}_A] \quad \text{for all } n \text{ and all } A \in \mathcal{F}_n,$$

then $(X_n)_{n \geq 0}$ is a *supermartingale*.

1.2 Natural filtration

Given a random process $X = (X_n)_{n \geq 0}$, its *natural filtration* $(\mathcal{F}_n^X)_{n \geq 0}$ is defined by

$$\begin{aligned}\mathcal{F}_n^X &= \sigma(X_0, X_1, \dots, X_n) \\ &= \{ \{ (X_0, X_1, \dots, X_n) \in B \} : B \in \mathcal{B}(\mathbb{R}^{n+1}) \}.\end{aligned}$$

This is the smallest filtration making X adapted.

Exercise. Show that a martingale X with respect to $(\mathcal{F}_n)_{n \geq 0}$ is also a martingale with respect to $(\mathcal{F}_n^X)_{n \geq 0}$.

1.3 Examples

Let $X_1, X_2, \dots$ be a sequence of independent integrable random variables. Set

$$\mathcal{F}_n = \mathcal{F}_n^X.$$

Define

$$S_0 = 0, \quad Z_0 = 1,$$

and

$$S_n = X_1 + \dots + X_n, \quad Z_n = \prod_{i=1}^n X_i.$$

- If $\mathbb{E}[X_n] = 0$ for all n , then $(S_n)_{n \geq 0}$ is a martingale.
- If $X_n \geq 0$ and $\mathbb{E}[X_n] = 1$ for all n , then $(Z_n)_{n \geq 0}$ is a martingale.

To see this, take $A \in \mathcal{F}_n$. Then

$$\begin{aligned}\mathbb{E}[S_{n+1} \mathbf{1}_A] &= \mathbb{E}[X_{n+1} \mathbf{1}_A] + \mathbb{E}[S_n \mathbf{1}_A] \\ &= \underbrace{\mathbb{E}[X_{n+1}] \mathbb{P}(A)}_{=0} + \mathbb{E}[S_n \mathbf{1}_A] \\ &= \mathbb{E}[S_n \mathbf{1}_A].\end{aligned}$$

Also,

$$\begin{aligned}\mathbb{E}[Z_{n+1} \mathbf{1}_A] &= \mathbb{E}[X_{n+1} Z_n \mathbf{1}_A] \\ &= \underbrace{\mathbb{E}[X_{n+1}]}_{=1} \mathbb{E}[Z_n \mathbf{1}_A] \\ &= \mathbb{E}[Z_n \mathbf{1}_A].\end{aligned}$$

This also shows integrability.

2 Conditional Probabilities and Expectation

For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

and

$$\mathbb{E}[X \mid B] = \frac{\mathbb{E}[X \mathbf{1}_B]}{\mathbb{P}(B)}.$$

2.1 Existence and uniqueness

Theorem. Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ -algebra. Let X be an integrable random variable. Then there exists a random variable Y such that:

(a) Y is $\mathcal{G}$ -measurable;

(b)

$$\mathbb{E}[X\mathbf{1}_A] = \mathbb{E}[Y\mathbf{1}_A] \quad \text{for all } A \in \mathcal{G}.$$

Moreover, if Y' is another random variable satisfying (a) and (b), then

$$Y = Y' \quad \text{a.s.}$$

Proof of uniqueness. Let X' be another integrable random variable with $X \leq X'$. Suppose Y' satisfies (a) and (b) with respect to X' .

Set

$$A = \{Y > Y'\}, \quad Z = (Y - Y')\mathbf{1}_A.$$

Then $A \in \mathcal{G}$ and $Z \geq 0$. Hence

$$\begin{aligned} \mathbb{E}[Z] &= \mathbb{E}[(Y - Y')\mathbf{1}_A] \\ &= \mathbb{E}[Y\mathbf{1}_A] - \mathbb{E}[Y'\mathbf{1}_A] \\ &= \mathbb{E}[X\mathbf{1}_A] - \mathbb{E}[X'\mathbf{1}_A] \\ &= \mathbb{E}[(X - X')\mathbf{1}_A] \leq 0. \end{aligned}$$

Therefore $Z = 0$ a.s., so

$$Y \leq Y' \quad \text{a.s.}$$

Taking $X' = X$, we also obtain $Y \geq Y'$ a.s. Thus

$$Y = Y' \quad \text{a.s.}$$

2.2 Notation

We write

$$Y = \mathbb{E}[X \mid \mathcal{G}] \quad \text{a.s.}$$

If $\mathcal{G} = \sigma(W)$, where W is a random variable, we write

$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X \mid W].$$

If $X = \mathbf{1}_A$, we write

$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{P}(A \mid \mathcal{G}).$$